

BinPulse: An IoT-Driven Software Approach for Intelligent Waste Collection in Smart Cities

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Problem Statement

Rapid urban growth in Indian cities has led to continuous increases in municipal solid waste. Current waste collection practices rely on fixed schedules that rarely reflect actual bin conditions, resulting in overflow incidents in high-usage locations whilst bins in low-demand areas are emptied unnecessarily.

These inefficiencies create sanitation challenges, increase fuel usage and operational expenditure, and contribute to environmental impact. Traditional methods fail to consider real-time bin status, leading to inefficient vehicle routing and resource allocation.

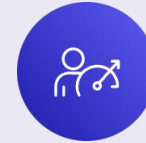


Research Objectives



Real-Time Monitoring

Develop cost-effective IoT sensors to continuously track garbage fill levels, weight accumulation, and hazardous gas presence



Optimise Collection

Replace fixed schedules with condition-based collection using priority indices and geographical proximity grouping



Enhance Safety

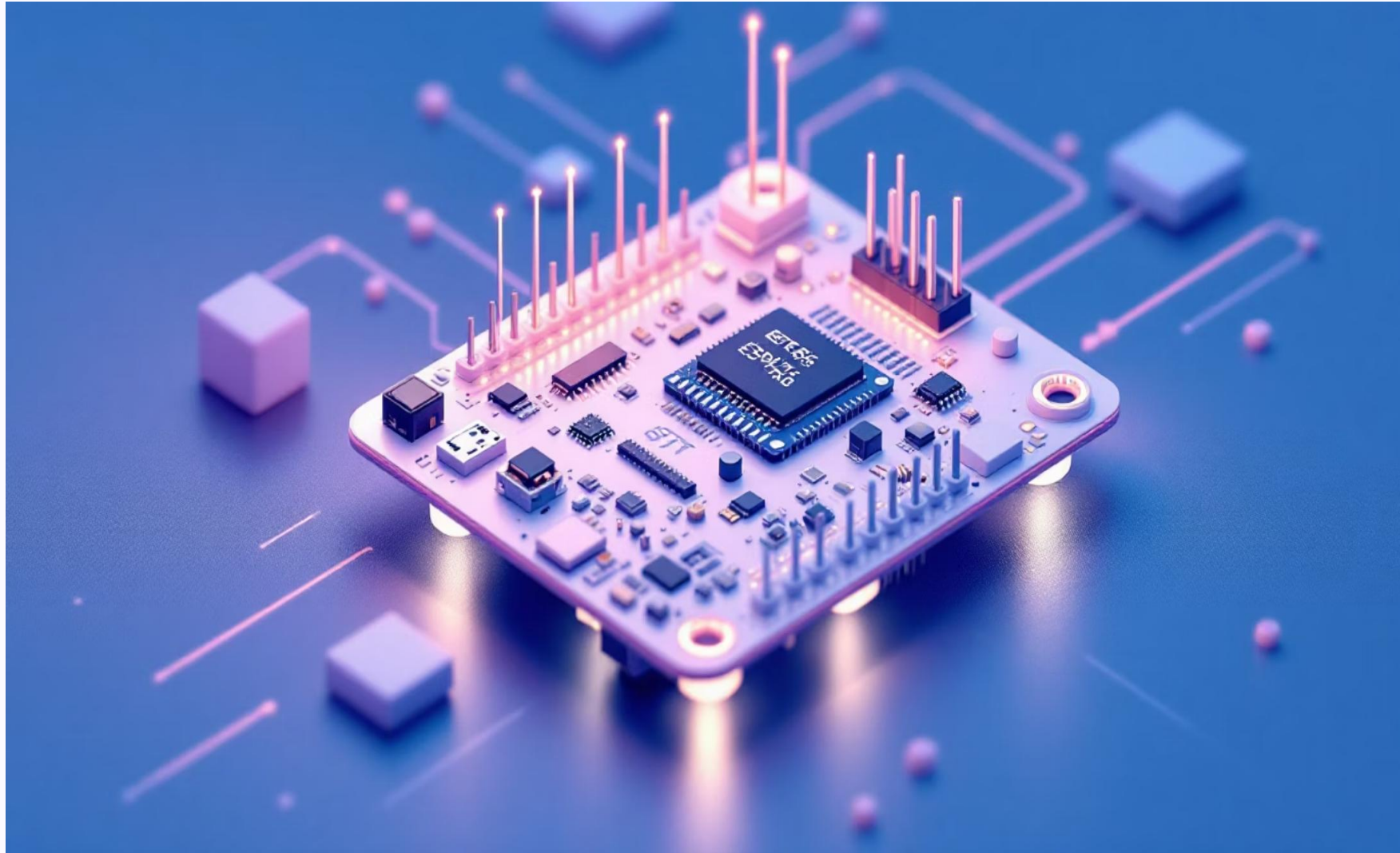
Detect flammable gases generated from waste decomposition to prevent fire hazards and emergency situations



Reduce Costs

Minimise unnecessary collection trips, fuel consumption, and operational expenses whilst maintaining service quality

Methodology Overview



System Architecture

The proposed methodology integrates low-cost IoT sensing hardware with centralised cloud-based data processing, enabling real-time monitoring and informed decision-making whilst maintaining simplicity and affordability.

Each garbage bin is equipped with sensors monitoring physical capacity and safety conditions. Sensor readings are transmitted to a Firebase-based backend using JSON format, where rule-based logic evaluates incoming data and determines service requirements.

Hardware Components

HC-SR04 Ultrasonic Sensor

Measures distance between sensor and waste surface to calculate fill level percentage using non-contact technique

Load Cell with HX711

Measures actual waste weight to account for compaction effects, providing more accurate bin condition assessment

MQ-2 Gas Sensor

Detects flammable and hazardous gases from waste decomposition, triggering immediate safety alerts when thresholds exceeded

ESP8266 NodeMCU

Primary data acquisition and communication unit, managing multiple bins with clustered deployment strategy





Priority Calculation Algorithm

The system employs a weighted priority index to determine collection order when multiple bins require attention simultaneously:

$$P = aF + bW_u + cG_r$$

F: Fill Level %

Garbage height percentage calculated from ultrasonic measurements

W_u: Weight Utilisation %

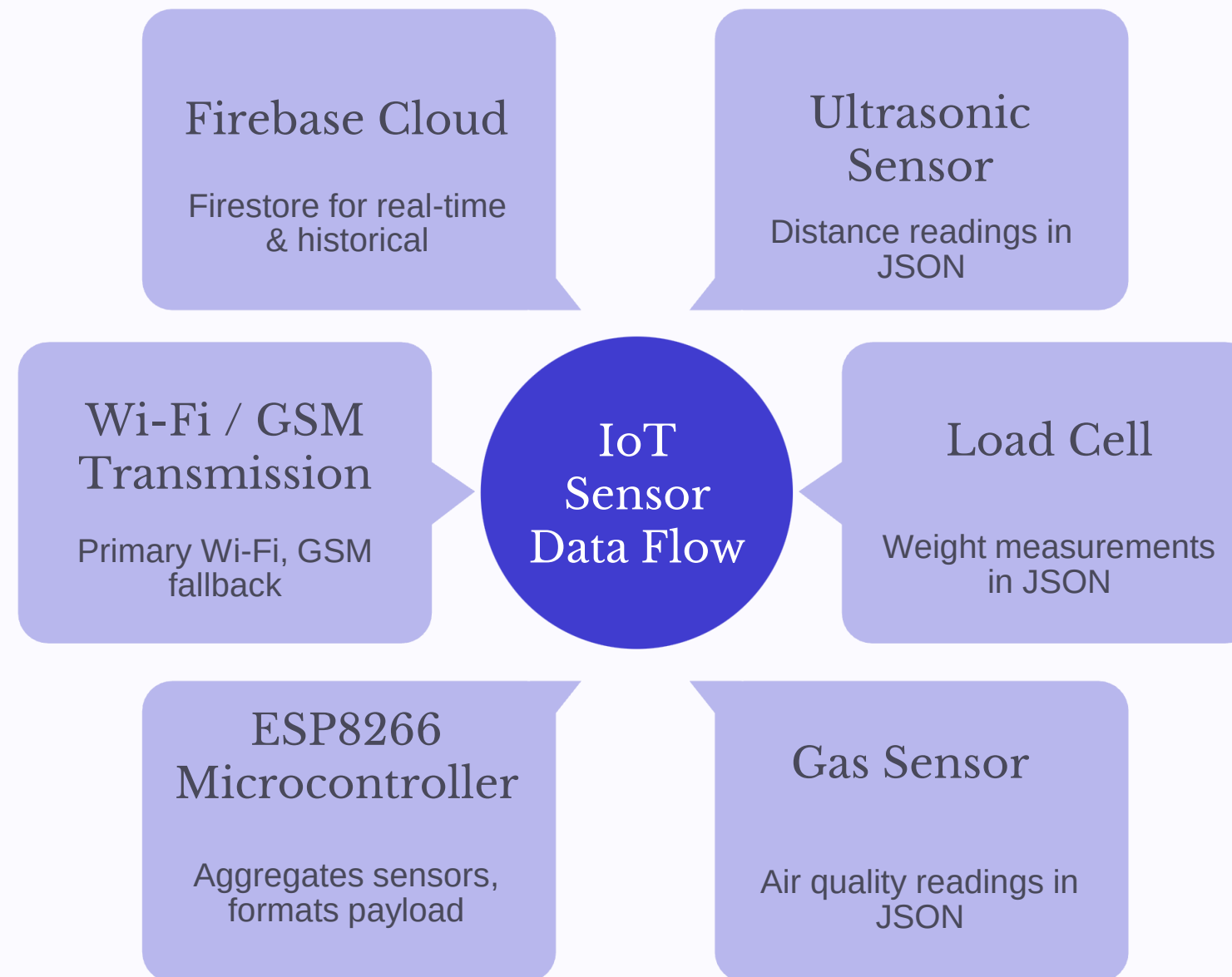
Measured waste weight divided by maximum allowable bin capacity

G_r: Gas Risk Index

Normalised safety risk derived from gas sensor readings with higher weighting

Higher priority indices receive earlier service, with gas risk emphasised through coefficient weighting.

Implementation Architecture



Sensor data are transmitted in lightweight JSON format using Wi-Fi as primary medium, with GSM as fallback. Firebase Firestore stores real-time status and historical time-series data. Municipal authorities monitor through web dashboard whilst collection drivers receive alerts via Flutter mobile application.

Key Results and Findings

40%

Reduced Collection
Trips

Elimination of unnecessary
collections through condition-
based monitoring

65%

Improved Overflow
Prevention

Early identification of bins
approaching full capacity

30%

Fuel Cost Reduction

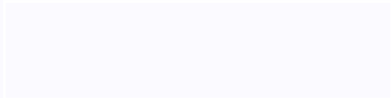
Optimised routing based on
priority and geographical
proximity

24/7

Continuous
Monitoring

Real-time safety alerts for
hazardous gas detection

The prototype demonstrated that data from low-cost hardware components effectively improved waste monitoring, safety awareness, and collection planning without complex algorithms or expensive infrastructure.



Future Enhancements

Predictive Analytics



Incorporate machine learning models using historical data to estimate waste generation trends and predict collection windows

GIS Integration



Combine with geographic information systems and real-time traffic data for dynamic route optimisation under varying conditions

Citizen Engagement



Add feedback mechanisms to enhance waste segregation practices and improve service responsiveness whilst maintaining scalability



Our Contribution

Cost-Effective Solution

Low-cost sensor configuration using readily available components with clustered deployment strategy reducing per-bin hardware requirements

Practical Deployment

Simple threshold-based rules and heuristic prioritisation sufficient for efficient collection without complex algorithms

Safety Integration

Comprehensive safety monitoring combining fill level, weight, and hazardous gas detection for complete waste management coverage

This work provides a practical pathway for implementing smart waste management in resource-constrained urban environments, delivering measurable economic and environmental benefits through intelligent, condition-based collection.

